Lab 4: Bash Scripting - Part 2

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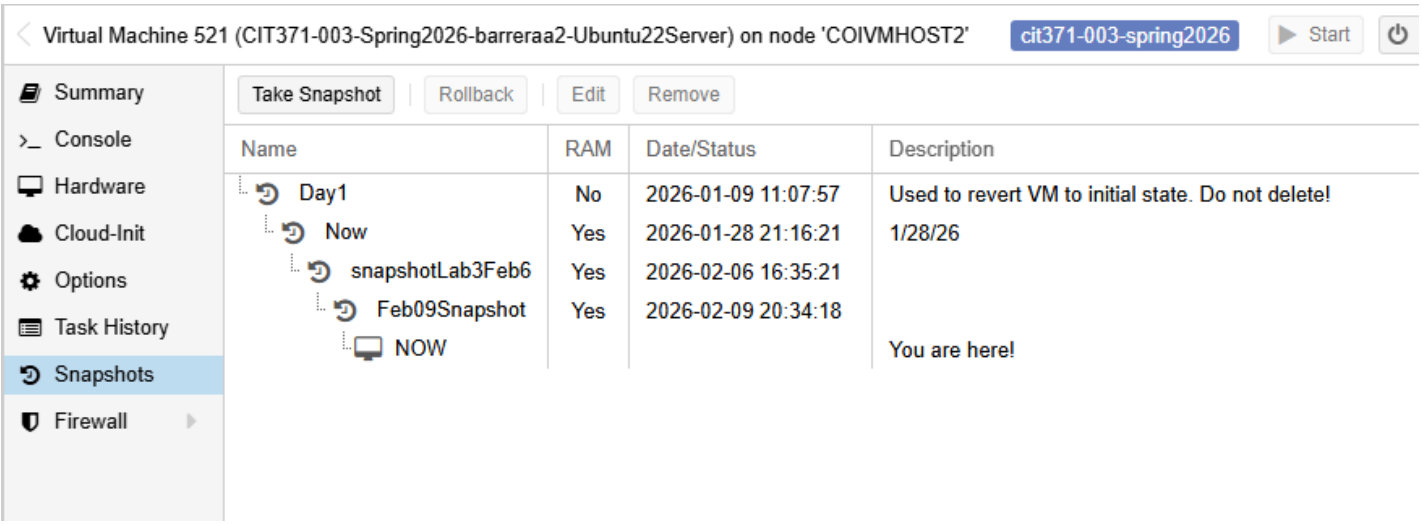
VM Snapshot(s)



Overview

In this Lab 04 Part 2 we will be working on a bash script that will help us get a text-based menu letting users from choosing between starting, stopping, restarting, enabling and checking the status of a system service . A user will be able to pick from this options below and depending on what they pick their will be a output on the choice.

The Menu should look like this..

Example:

-----SERVICE-SCRIPT-MENU-----

1. Start a Service.
2. Stop a service.
3. Restart a service.
4. Enable a service.
5. Check a service's status.
6. Clear screen.
7. Exit script.

Please choose an option by entering its number (i.e., 1,2):

In this script if the user picks the options 1-6 then the script will reprint ("or return to") the menu after they selected a number so they could pick another option if they want after the previous one is completed.

Okay, Now we will create a script using bash, First we need to create the script in the command line using the command below to have a name for the bash script as well:

```
nano service_menu.sh
```

Then we will get the memo windows to write our script.

First we will start adding what is necessary for the script to be executable:

```
#!/bin/bash
```

Then we will also below start adding the menu by putting all this information as command:

```
# Function to print the menu
print_menu() {
    echo "-----SERVICE-SCRIPT-MENU-----"
    echo "1. Start a service."
```

```
    echo "2. Stop a service."
    echo "3. Restart a service."
    echo "4. Enable a service."
    echo "5. Check a service’s status."
    echo "6. Clear screen."
    echo "7. Exit script."
    echo -n "Please choose an option by entering its number (i.e., 1, 2): "
}
```

After we created the menu, we will start a loop with conditions to keep asking the user if they want to pick another options after their previous option has been completed with this code:

- From 1-6 we will make sure that reprint or ("return to") the menu and allow the user to select another options after they have the previous one completed.
- For 1-4 we should let the user know if the option was successful or not before doing this.
- If the user wants to quit, then the only option they can pick is number 7. or terminate it manually typing Ctrl + C shortcut.
- For options 1-5 we will be asking the user for specify the service and then use the proper systemctl command to performance the the stated operation to it.
- Options 1-4 should verify the success of the systemctl command used by checking if it.
- • Unlike the first four, option 5 should just request a service from the user and use the systemctl command to check and print out the service’s status.
- • Option 6 should use the clear command to clear the screen.
- • Option 7 should terminate the script using a bash command for doing so.

With all this information we will write the loop now with this command below one number by the other:

```
# Main loop
while true; do
   print_menu
   read choice

   case $choice in
   1) echo -n "Enter the service you want to start: "
      read service
      sudo systemctl start "$service"
      if [ $? -eq 0 ]; then
          echo "SUCCESS: $service started successfully."
      else
          echo "FAILURE: Could not start $service."
      fi
      echo
```

```
      ;;
    2)
      echo -n "Enter the service you want to stop: "
      read service
      sudo systemctl stop "$service"
      if [ $? -eq 0 ]; then
           echo "SUCCESS: $service stopped successfully."
      else echo "FAILURE: Could not stop $service."
      fi
      echo
      ;;

    3)
      echo -n "Enter the service you want to restart: "
      read service
        sudo systemctl restart "$service"
      if [ $? -eq 0 ]; then
        echo "SUCCESS: $service restarted successfully."
      else
        echo "FAILURE: Could not restart $service."
      fi
        echo
        ;;

    4)
       echo -n "Enter the service you want to enable: "
       read service
       sudo systemctl enable "$service"
       if [ $? -eq 0 ]; then
          echo "SUCCESS: $service enabled successfully."
       else
          echo "FAILURE: Could not enable $service."
       fi
       echo
       ;;

    5)
       echo -n "Enter the service whose status you want to check: "
       read service
       echo
       sudo systemctl status "$service"
       echo
       ;;
    6)
       clear
       ;;
    7)
       echo "Exiting script..."
```

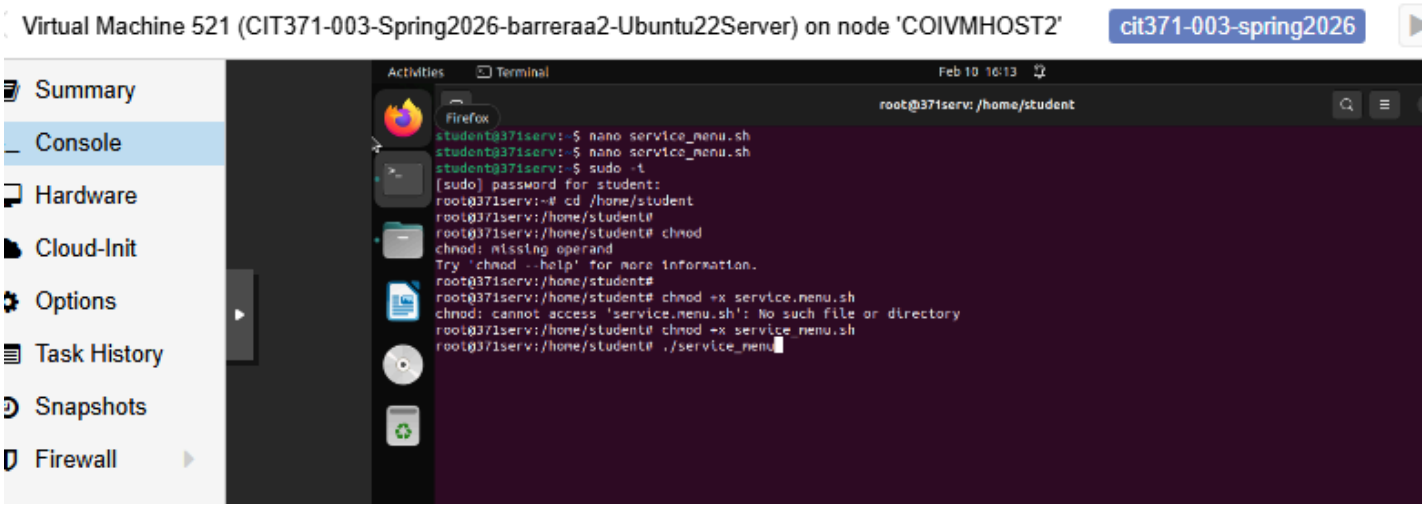
```
        exit 0
        ;;
    *)
        echo "Invalid option. Please choose a number from 1 to 7."
        echo
        ;;

  esac
done
```

Now that we have the bash script ready to be execute it, we will need to run it using “sudo -i” to sign into root before running the script. Most of the systemctl commands will not work without elevated permissions.

Let's run this command now:

```
sudo -i
cd /home/student
chmod +x service_menu.sh
./service_menu.sh
```

It will ask for a password and then you will be in the directory where the script is, so we could run it.

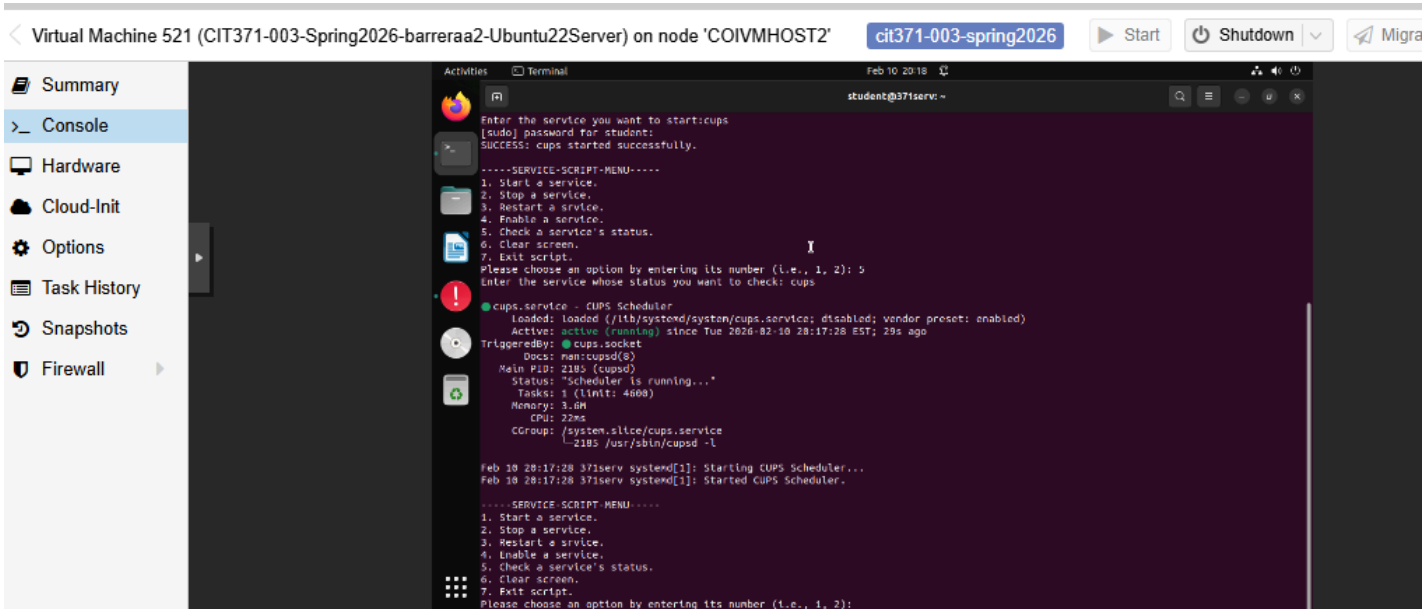


Now that we created the script and is ready to run we will test it by running these commands:

```
cd /home/student
chmod +x service_menu.sh
./service_menu.sh
```

Now we will pick the options to start the “cups” service. We will type 1.

Then we will ask for the cups status to make sure it has started by picking number 5 as it shows below..



References

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